

DIPENDRA SHARMA

Aspiring Backend Engineer | Building Scalable, Reliable System

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Summary

I'm a backend-focused developer who enjoys building reliable systems and learning new technologies through real-world projects. I've built and deployed **REST APIs**, worked with relational databases, and I'm currently learning **GraphQL** to depth my API design skills. I have a strong foundation in **databases, data structures, algorithms, operating systems**, and object-oriented programming. I approach problems from first principles, tracing issues through system behavior. I take ownership of features end-to-end and enjoy working across the backend stack. I'm currently exploring **Redis**, caching strategies, system design, load balancing, and distributed systems to build scalable and maintainable software.

Technical Skills

Languages: JavaScript, TypeScript, Java, C++, SQL

Backend: Node.js, Express.js, REST APIs, JWT, RBAC, Zod, Jest, Supertest

Databases: PostgreSQL (indexing, CTEs, EXPLAIN ANALYZE, connection pooling), MongoDB, MySQL, Supabase

DevOps & Linux: Docker, Git, GitHub Actions, Vercel, Render, AWS (EC2, S3, IAM), Linux, bash, cron, Apache Bench

Security & Performance: Rate limiting, bcrypt, JWT, query optimisation, load testing, input validation

Projects

Yeti Jobs | [GitHub](#) | [Live](#)

Jan 2026 – Apr 2026

- Built **50+ REST APIs** with RBAC across 3 user roles and **10 normalized PostgreSQL tables** — wrote real SQL with JOINS, CTEs, and composite indexes. Query time dropped from **7ms to 5ms (30%)** per EXPLAIN ANALYZE. Connection pooling for **20 concurrent DB connections**.
- Debugged **10+ production issues** to root cause — PostgreSQL pool leaks, cold start latency (**50s → 2s**), duplicate cron jobs. Traced stack traces and read query plans to find what broke and why.
- Load-tested with Apache Bench: **1,000 requests at 100 concurrent users → 67 req/sec, p95 1,929ms, 0 failures** on a single free-tier Render instance.
- Wrote **28+ integration tests** (Jest + Supertest) covering 11 critical routes. Cron job every 10 minutes eliminated cold starts, sustaining **99.5% uptime** over 3 months.
- Multi-stage Docker builds (Alpine) reduced backend image by **74% (1.99GB → 520MB)**. Deployed on Render with GitHub Actions CI/CD; implemented Express rate limiting to **mitigate** resource abuse and automated server keep-alive.
- **Tech: PostgreSQL, Node.js, Express, JavaScript, TypeScript, Docker, JWT, Jest, Supertest, GitHub Actions**

Daigo | [GitHub](#)

May 2026 – Present

- Building a two-sided marketplace riders post requests, drivers accept which solved **cold-start bootstrapping, multi-hop relay matching** (chaining two drivers), and **geospatial route overlap** using raw PostGIS queries via Prisma \$queryRaw.
- Designed a **pricing engine** with per-zone multipliers, time-of-day bands, and demand multipliers — two hard caps (per-km and absolute) checked against the **rounded value, not raw float**, removing a floating pricing bugs
- Implemented **real-time accept flow** using WebSocket + Redis pub/sub for cross-instance delivery — 75-second countdown, first-accept-wins concurrency, driver alerts. Redis handles in-trip chat, rate limiting, and cached trust scores.
- build **failure-resistant backend** with idempotency keys for payment retries, webhook verification for Razorpay mock, atomic seat release, and **background job processing** via Upstash QStash for exports, verification, and alerts.
- Integrated **signed-URL direct-to-storage uploads** (Supabase) to bypass backend handling — validating type/size server-side, clients upload directly, reducing API load and memory footprint.
- **Tech: Next.js, TypeScript, Prisma, PostgreSQL + PostGIS, Redis, QStash, Stadia Maps, Razorpay**

Achievements

- Selected for **GirlScript Summer of Code 2026**.
- Winner of **CodeaThon 7.0** (cash prize, 120+ participants).
- **313 LeetCode** problems solved | **20+ GitHub repos** | **1000+ commits**.
- Shortlisted for **Round 2 – Polaris Fellowship 2026** (selected from thousands across 300+ cities).
- **CEED Member** (Entrepreneurship Program), Chitkara University.

Certifications

- **AWS Educate Introduction to Cloud 101** – EC2, S3, IAM fundamentals
- **Web Development Masterclass** – HTML/CSS, JavaScript, Node.js, REST APIs

Education

Chitkara University

Aug 2024 – Expected 2027

Bachelor of Computer Applications (BCA) | CGPA: 9.44/10

Himachal Pradesh, India

- Coursework: DSA, Operating Systems, DBMS, Software Engineering, Computer Networking, Web Performance